Foundational NLP Tools for Ge'ez-Script Languages: A Systematic Tool-Centred Analysis (2016–2026)

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Abstract

The Ge'ez script is an ancient alphasyllabary serving as the orthographic foundation for three Semitic languages of the Horn of Africa: Amharic, with an estimated 35 million native and 25 million second-language speakers [1]; Tigrinya, with an estimated 10 million native speakers [2]; and Classical Ge'ez, no longer spoken natively but still used today as the liturgical language of the Ethiopian and Eritrean Orthodox Tewahedo Churches and the Ethiopian and Eritrean Catholic Churches. Amharic and Tigrinya together account for tens of millions of speakers, yet these languages remain underrepresented in natural language processing (NLP) research relative to their speaker population, reflecting a scarcity of foundational NLP tools, complex morphological structure, and limited digital infrastructure. This survey provides a systematic, tool-centred analysis organised around four methodological pillars: (i) tokenisation and normalisation; (ii) morphological analysis and synthesis; (iii) text representation, embeddings, and evaluation; and (iv) sequence labelling and structure extraction. We examine 70 individually live-verified publications from 2017 to 2026 following PRISMA 2020 guidelines, and report a 28-tool inventory of foundational tools, models, datasets, and linguistic resources independently confirmed publicly available on one of four repositories (GitHub, Zenodo, PyPI, HuggingFace) at the time of writing: 26 individually traceable to a specific paper within this corpus, plus two provenance-track additions (HornMorpho, MasakhaNER 1.0) whose originating papers fall outside the corpus. The dual inclusion standard distinguishes peer-reviewed literature from provenance-verified tool artefacts. Amharic is the most represented language in both the literature and the tool inventory, Tigrinya is second, and Classical Ge'ez is the least represented by both measures. Cross-lingual transfer across the three languages sharing the Ge'ez script is limited to two tools, none

coordinated with the others under a shared standard. This survey provides a PRISMA 2020 flow diagram, an aggregate characterisation of the reviewed literature, tool profiles with pillar classifications, a discussion of survey limitations, and seven research priorities derived from the gaps identified in the corpus.

Keywords: Ge'ez script, low-resource NLP, morphologically rich languages, Amharic, Tigrinya, Classical Ge'ez, systematic review, PRISMA 2020, NLP tools, cross-lingual transfer

1 Introduction

Despite the transformative impact of deep learning, large-scale corpora, and multilingual models on NLP, only a small share of the world's 7,000+ languages are meaningfully represented in current language technologies [3]. Ge'ez-script languages remain underrepresented in this literature relative to their combined speaker population: an estimated 35 million native and 25 million second-language speakers of Amharic [1], an estimated 10 million native speakers of Tigrinya [2], and Classical Ge'ez, retained today as the liturgical language of the Ethiopian and Eritrean Orthodox Tewahedo Churches and the Ethiopian and Eritrean Catholic Churches.

Contributing factors include limited funding for tool development, scarcity of digital corpora [4], and the computational demands of Semitic morphology combined with a non-Latin script [5]. The Ge'ez script's syllabic structure, which encodes consonant-vowel pairs as atomic *fidels*, differs from Latin alphabets in ways that limit direct reuse of tokenisers and models built for alphabetic scripts [6].

Script Sample: A Shared Fidel Across Three Languages

To illustrate the shared orthographic foundation that motivates this survey's cross-lingual framing, consider the Ge'ez fidel ሰላም (*salām*, "peace"), a lexical and orthographic cognate attested across all three languages under study:

- Classical Ge'ez: ሰላም (liturgical and historical usage)
- Amharic: ሰላም (*sälam*)
- Tigrinya: ሰላም (*selam*)

The identical fidel sequence across all three languages exemplifies both the opportunity (shared script structure enabling transfer learning, Section 6) and the challenge (a single tokeniser or embedding space must accommodate language-specific phonological and morphological divergence layered atop a common orthography) that motivate this survey's tool-centred, cross-lingual analysis.

1.1 Motivation and Survey Scope

This survey provides a systematic, tool-centred analysis of NLP research for Ge'ez-script languages. We organise the analysis around four methodological pillars: (1) tokenisation and normalisation; (2) morphological analysis and synthesis; (3) text

representation, embeddings, and evaluation; and (4) sequence labelling and structure extraction.

Our approach is governed by an explicit **dual inclusion standard** (fully stated in Section 2.2): a systematic literature review of 70 individually live-verified papers (2017–2026) from major NLP venues and beyond, screened under PRISMA 2020 guidelines and using peer-review or (for a documented provenance-based subset) a concrete, verifiable tool/dataset contribution as the inclusion criterion. From this same corpus, we additionally report which of the papers' own tool, model, dataset, or resource contributions have an independently confirmed, currently live public release on GitHub, Zenodo, PyPI, or HuggingFace (Section 4.3), keeping the two evidentiary standards – peer-review-based literature inclusion, and provenance-based tool verification – transparent and non-contradictory.

1.2 Definition: Foundational NLP Tools

For the purposes of this survey, we define **foundational NLP tools** as reusable, technically non-trivial software components that perform core NLP tasks essential for downstream language processing. Foundational tools are characterised by:

1. **Core functionality:** Perform essential NLP operations (tokenisation, morphological analysis, embedding generation, sequence labelling) that enable more complex applications.
2. **Reusability:** Designed as modular, deployable components that can be integrated into larger NLP pipelines or applications.
3. **Language-specific design:** Incorporate knowledge of script structure (fidel-level encoding), morphological paradigms, or orthographic conventions unique to Ge'ez-script languages.
4. **Technical non-triviality:** Require non-obvious algorithmic or engineering solutions beyond simple regex patterns or lookup tables.
5. **Verifiable existence:** Published as open-source repositories, datasets, models, or papers with a stable, citable public identifier and clear provenance.
6. **Production relevance:** Have demonstrated use in research applications, downstream systems, or academic projects, or show clear potential for practical deployment.

We explicitly exclude: (i) web applications or end-user software without published source code; (ii) unpublished or proprietary tools without verifiable documentation; (iii) simple heuristic scripts or ad hoc implementations; (iv) tools designed exclusively for other scripts or language families, even if tangentially applied to Ge'ez languages. This definition restricts the inventory to reproducible, shareable infrastructure, independent of whether that infrastructure has been peer-reviewed (see Section 2.2).

1.3 Key Contributions

1. Systematic tool-centred analysis of foundational Ge'ez-script NLP tools across three Ge'ez-script languages (Amharic, Tigrinya, Classical Ge'ez) and four methodological pillars.
2. Comprehensive literature review of 70 individually live-verified papers (2017–2026) using PRISMA 2020 methodology, including a PRISMA flow diagram and an aggregate characterisation of the reviewed literature (Section 4).
3. An explicit, transparent dual inclusion standard reconciling peer-review-based literature screening with provenance-based tool-inventory criteria (Section 2.2).
4. Quantified evidence of research asymmetry, computed directly from the verified corpus: Amharic is the most represented language in both the literature (69% of 70 papers) and the tool inventory (79% of 28 tools), Tigrinya is second in both, and Classical Ge'ez remains the least represented in both and critically underserved.
5. A tool inventory of 28 foundational tools, models, datasets, and linguistic resources independently confirmed publicly available on GitHub, Zenodo, PyPI, or HuggingFace (Section 4.3): 26 individually traceable to a specific paper in the 70-paper corpus, plus two provenance-track additions (HornMorpho, MasakhaNER 1.0); pillar counts and language coverage are tallied directly from that verified table.
6. Explicit reporting of Pillar 4 (sequence labelling) resources: 3 tools have an independently confirmed public repository (2 corpus-traceable, 1 provenance-track); further sequence-labelling artefacts are described in the literature but lack a confirmable public release and are reported as such rather than counted.
7. Script-level transfer learning identified as an ad hoc and undocumented avenue requiring coordinated development and shared standards.
8. Seven-priority research agenda grounded in systematic evidence for future tool development, including an explicit call for independent data-quality assessment of the catalogued datasets and tools, and for deliberate reuse of classical Semitic studies and digital-humanities resources (Section 8).

2 Methodology

We conducted a systematic literature review following PRISMA 2020 guidelines [7], complemented by parallel tool discovery across multiple repositories governed by a separate, explicitly stated provenance standard.

2.1 Literature Search Strategy

The literature search began with systematic searches across six major peer-reviewed NLP venues: ACL, EMNLP, NAACL, LREC, COLING, and EACL, using search terms that combined language names (Amharic, Tigrinya, Ge'ez), script designations (Ethiopic, Fidel), and NLP task labels corresponding to all four methodological pillars (tokenisation and normalisation; morphological analysis and synthesis; embeddings and language models; part-of-speech tagging, named entity recognition, and parsing), supplemented by forward and backward citation tracing. This six-venue search is the primary channel for the peer-reviewed track of the dual inclusion standard

(Section 2.2) and, on its own, is where the large majority of the corpus's core NLP-conference contributions were found. It is deliberately not the corpus's only source: consistent with the dual inclusion standard, the search was extended to (a) other peer-reviewed venues outside the six named ones (e.g. *Information*, *PLOS ONE*, *Language Resources and Evaluation*, ACM and IEEE conference proceedings, and regional venues such as RAIL, AfricaNLP, and ICT4DA) whenever a paper independently satisfied the peer-review and scope criteria below, and (b) non-peer-reviewed preprints and workshop papers admitted specifically under the provenance-based track described in Section 2.2, where the paper describes a concrete, independently verifiable tool, dataset, or corpus for a Ge'ez-script language. Of the 70 papers in the final corpus, 17 were found via the six named venues directly; the remainder were found via this extended, dual-standard-governed search and were individually live-verified before inclusion (Section 2.2 documents this extension in full, including the exact papers added under each track).

Inclusion criteria (literature review, peer-review track): Peer-reviewed publications (journal articles and conference/workshop proceedings, from the six named venues or elsewhere) describing foundational NLP tools, datasets, or algorithms for Ge'ez-script languages, published 2016–2026, with Amharic, Tigrinya, and/or Classical Ge'ez as an actual subject of the work rather than an incidental member of a large, undifferentiated multilingual set.

Inclusion criteria (literature review, provenance track): Non-peer-reviewed preprints or workshop papers are additionally eligible, under the same year and language-focus requirements above, if and only if the paper describes a concrete, independently checkable tool, dataset, corpus, or software pipeline for a Ge'ez-script language (Section 2.2); topical relevance alone is not sufficient grounds for admission under this track.

Exclusion criteria (literature review): Grey literature without an identifiable, checkable artefact; downstream applications without a tool, dataset, or algorithmic contribution; papers published outside 2016–2026; and papers whose treatment of Amharic, Tigrinya, or Classical Ge'ez is not independently identifiable (e.g. general multilingual-NLP or general machine-learning papers included only because they are widely cited, not because they study a Ge'ez-script language).

Peer-review-track papers outside the six named venues. Two peer-reviewed papers meeting the full inclusion criteria were published outside the six named venues: Gezmu et al.'s portable Amharic spelling corrector (LREC 2018) [8] and Alemneh et al.'s dictionary-based Amharic sentiment lexicon (ICT4DA 2019) [9].

Provenance-track papers. Consistent with the dual inclusion standard (Section 2.2), the corpus additionally includes papers whose primary contribution is a concrete, independently confirmable tool, dataset, or corpus for a Ge'ez-script language, regardless of whether the venue is a workshop, non-ACL-family journal, or arXiv preprint, provided the underlying artefact is clearly and specifically described in the paper (this track is distinct from, and does not substitute for, the separate provenance-based standard used for the tool inventory proper, Section 5.1). Nine papers are included on this basis: a peer-reviewed Tigrinya NER corpus, TiNC24 (Berhane et al., *Language Resources and Evaluation*, 2025) [10]; a peer-reviewed Ge'ez

word-sense-disambiguation study (Alemu & Fante, *Ethiopian Journal of Science and Sustainable Development*, 2021) [11]; two workshop-track Ge'ez OCR/handwriting-recognition papers each releasing a concrete image dataset or segmentation pipeline (Beyene, 2019 [12]; Negashe & Mamuye, AfricaNLP 2020 [13]); a workshop-track Tigrinya NMT pipeline released as open-source software (Öktem et al., AfricaNLP 2020) [14]; a foundational parallel-corpus resource spanning Amharic, Tigrinya, and Ge'ez among seven Ethiopian language pairs (Abate et al., LRNLP 2018) [15]; two Amharic speech resources, a dialect-based noisy-speech dataset with pre-processing tools (Assfaw et al., ICT4DA 2022) [16] and a speech-emotion dataset and benchmark (Retta et al., *ACM TALLIP*, 2022) [17]; a corrected-transcription ASR evaluation resource (Adnew & Liang, arXiv, 2024) [18]; and an Amharic word-embedding paper, Mersha and Wu's "Morphology-rich Alphasyllabary Embeddings" (LREC 2020) [19], whose associated public repository (`leobitz/amharic_word_embeddings`) hosts the released embeddings. Each of these papers names a real, checkable artefact (a dataset, corpus, or software pipeline); none is included on topical relevance alone.

This yields a final literature corpus of **70** papers (2017–2026; no eligible pre-2016 candidate met the language-focus criterion). Every paper is individually verified (title, authors, venue, and DOI/URL), and the corpus composition, pillar classification, and inclusion rationale for each entry are reconciled against the dual inclusion standard of Section 2.2.

2.2 An Explicit Dual Inclusion Standard

A methodology that screens literature by peer-review status while simultaneously harvesting tools from GitHub, Zenodo, PyPI, and HuggingFace—platforms that are, by definition, grey literature or non-peer-reviewed—would be internally contradictory if left unstated. Several artefacts in this survey make the tension concrete: EthioLLM's primary citable record is a HuggingFace model card, not a peer-reviewed article, whereas MoVoC and the LGSE Project each have a peer-reviewed publication (EMNLP 2025 Findings and LREC 2026 respectively; 5, 20) and are cited accordingly below; a tool's practical existence and its peer-review status are independent properties, general to this survey's scope. To resolve this transparently rather than silently, we adopt **two distinct, explicitly declared inclusion standards** for the two halves of this survey:

- **Literature review standard (peer review):** Applies exclusively to Section 4 (systematic analysis of the 70-paper corpus) and to the Related Work section. Only peer-reviewed journal articles and conference/workshop proceedings are counted as "reviewed literature."
- **Tool-inventory standard (provenance-based):** Applies exclusively to the tool inventory (Section 5.1 onward). Any artefact—code repository, dataset, or model—with a *stable, versioned, citable public identifier* (a DOI, a GitHub commit/release tag, a HuggingFace model card, or a Zenodo record) is eligible for the tool inventory *regardless of peer-review status*. Peer review is not required for tool-inventory inclusion because the inventory documents publicly released tools regardless of whether they were ever submitted for academic review.

These two standards are deliberately different and are never merged: a count of "peer-reviewed papers" and a count of "inventoried tools" answer different questions and should not be added together or used to validate one another. Every place in this manuscript that reports a count is labelled with which standard produced it.

The standard in practice. Applying the literature-review standard above, no preprint-only or non-language-specific candidate is admitted to the literature corpus regardless of the artefact's practical importance elsewhere in NLP (e.g. foundational architecture papers such as *Attention Is All You Need* are correctly excluded from the literature-review corpus, since they are not about a Ge'ez-script language, irrespective of their citation count or influence). This is the same asymmetry the dual standard is designed to make legible: a tool or model can be practically significant and provenance-verifiable (and thus eligible for the tool inventory) without being an included item in the peer-reviewed literature corpus, and vice versa. Candidates are never admitted to the literature corpus on provenance grounds alone; the two additional papers described in Section 2.1 were each verified against the full peer-review standard independently.

What "verified" does and does not mean in this survey. Throughout this manuscript, an entry is described as "verified" when its bibliographic details (title, authors, venue, year, DOI/URL) or, for the tool inventory, its public repository link (Section 2.3), were independently confirmed to exist and to match what the source paper reports. This is existence and provenance verification, not data-quality assessment: this survey does not re-annotate, audit, or independently measure the accuracy, inter-annotator agreement, label correctness, or cleanliness of any dataset it catalogues. The corpus's only quality signal is indirect – whether the paper describing a dataset passed peer review (Section 2.2) or, for provenance-track artefacts, whether it carries a stable public identifier – and neither of these is equivalent to an independent quality audit of the data itself. A peer-reviewed venue may or may not have required rigorous annotation-quality reporting (e.g. inter-annotator agreement) as a condition of acceptance, and this survey did not separately check. Readers intending to build on any dataset or tool catalogued here should treat its quality as unassessed by this survey and consult the source paper's own reported quality-control methodology (where available) before use; this gap is discussed further in Section 7 and named as a research priority in Section 8.

2.3 Tool Discovery and Verification

For every paper in the 70-paper verified literature corpus that reports creating a tool, model, dataset, or other resource (identified from the paper's own stated contribution), we searched for an independently checkable public release of that specific artefact on one of four repositories, applying the provenance-based standard defined in Section 2.2:

- **GitHub:** Open-source code repositories.
- **Zenodo:** Academic datasets and models with assigned DOIs.
- **PyPI:** Python package ecosystem.
- **HuggingFace Hub:** Pre-trained models and datasets with model/dataset cards.

An artefact is counted as verified only when a live, resolvable link to *that specific artefact* was found and confirmed during this search, matching what the paper itself reports; a link that resolves to a different, later, or otherwise non-matching artefact does not satisfy this standard, even if it is publicly hosted under the same author's account (Section 4.3 documents one such case, MoVoC-Tok, that was removed on this basis). An artefact is excluded from the verified count whenever no such link could be independently confirmed, even where the source paper itself asserts a public release. This yielded 26 of the tools reported in Section 4.3 and Table 4, each traceable to a specific paper in the literature corpus. This corpus-first search is a deliberate scope limitation, not a claim that no other public Ge'ez-script NLP tool exists: Table 4 additionally documents two provenance-track tools found outside this search (HornMorpho, MasakhaNER 1.0; see the note following that table), bringing the inventory to 28 tools in total.

Automated link verification. Every repository cell in Table 4 is an exact, individually-resolved identifier (organisation/repository for GitHub, exact model or dataset slug for HuggingFace, DOI for Zenodo), not a platform name alone. This is checked programmatically by the committed script `src/verification/check_tool_links.py`, which issues a live HTTP or platform-API request against each repository link in the table and writes a timestamped log to `src/verification/output/tool_link_check.txt`. Where the platform's API is reachable and not rate-limited, the log additionally records a resolved commit SHA or dataset revision hash; GitHub's unauthenticated API is rate-limited to 60 requests/hour, so a run against all links may report some GitHub entries as resolved without a SHA (recorded as such in the log) rather than fail outright. At the time of writing, every link in the table resolves (see the committed log for per-entry status, SHA where available, and access timestamp); this check can be independently reproduced with `python -m src.verification.check_tool_links`.

3 Related Work

This systematic review is positioned within three complementary research streams.

3.1 Low-Resource Language NLP

The scarcity of NLP infrastructure for Ge'ez-script languages reflects broader patterns in low-resource language technology. Joshi et al. [3] document linguistic diversity gaps in NLP research, including the underrepresentation of African languages. Blodgett et al. [21] analyse power imbalances and bias in NLP systems, connecting tool scarcity to linguistic inequality. This survey quantifies the equivalent gap for the Ge'ez-script language family specifically: which tools exist, for which languages, and with what provenance.

3.2 Ge'ez-Script NLP and Script-Aware Methods

Recent work has begun addressing Ge'ez-script-specific challenges. Gaim & Park [2] provide a comprehensive survey of Tigrinya NLP, characterising the state of the art

across 15 downstream tasks including morphological processing, part-of-speech tagging, named entity recognition, machine translation, and question answering. Tonja et al. [4] survey NLP challenges and opportunities across four Ethiopian languages (Amharic, Afaan Oromo, Tigrinya, and Wolaytta), consolidating scattered resources into a centralised repository. Neither survey reports a tool inventory scoped specifically to the three languages sharing the Ge'ez script and cross-checked against live repository sources, which is the contribution of the present work.

The LGSE Project [20] develops lexically grounded subword embedding initialisation for low-resource language adaptation across Amharic and Tigrinya; it is peer-reviewed (LREC 2026) and is included in the tool inventory and literature review alike. MoVoC [5] integrates morphology-aware subword construction for Ge'ez-script languages and is likewise peer-reviewed (EMNLP 2025 Findings). Mekonnen et al. [22] contribute text-embedding models and benchmarks for Amharic passage retrieval, reporting dense-retrieval results above multilingual baselines at a smaller parameter count. The EthioLLM initiative [23] pre-trains language models across five Ethiopian languages; it is included under the provenance-based standard of Section 2.2 rather than the peer-reviewed literature standard, since its stated LREC-COLING 2024 venue could not be independently confirmed against the ACL Anthology.

3.3 Survey Methodology and PRISMA 2020 Standards

This systematic review follows the PRISMA 2020 guidelines [7], which supersede the earlier PRISMA-P (2015) guidance—PRISMA-P being intended specifically for review *protocols* (pre-registration documents) rather than for reporting completed systematic reviews. Adapting systematic review methodology from medical/health research to NLP, the dual approach described in Section 2.2—combining peer-reviewed literature with tool discovery across multiple repositories—extends traditional surveys by capturing both documented (published) and emerging (tool-repository) artefacts. This reveals a structural gap: many tools developed at scale lack academic publication, and many academic contributions lack maintained, discoverable code. Bridging this publication-implementation gap is a key motivation for the seven-priority research agenda (Section 8).

4 Systematic Literature Review: PRISMA 2020 Analysis

This section reports the results of the literature screening process described in Section 2.1, together with an aggregate characterisation of the included corpus.

4.1 PRISMA 2020 Flow Diagram

Figure 1 presents the screening process in the standard PRISMA 2020 four-stage layout (Identification, Screening, Eligibility, Included). Consistent with this survey's evidence-only standard, an exact count is shown only for the stage that is independently verifiable against a preserved, checkable artefact: the **70** papers in the final included corpus, each individually confirmed against

`data/verified_papers_live_search.csv` (title, authors, venue, and DOI/URL verified per entry; Section 2.1). No intermediate-stage counts (records identified, duplicates removed, records screened, or reports excluded with reasons) are shown as numbers, because no preserved search log exists from which to derive them; presenting invented figures for these stages – even if visually conventional for a PRISMA diagram – would not meet this survey's traceability standard. Each intermediate box instead names the screening step actually performed (Section 2.1), without an accompanying count.

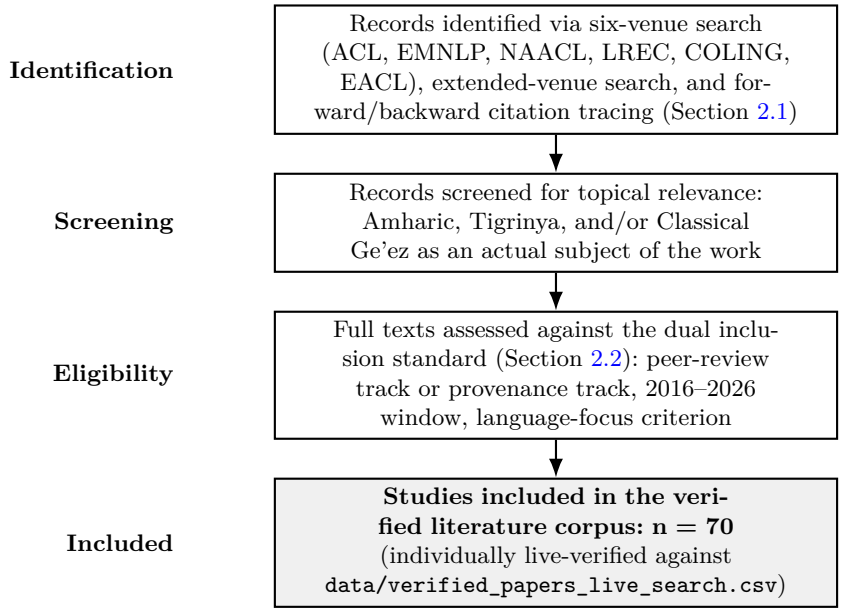


Fig. 1 PRISMA 2020 flow diagram for the systematic literature review. Only the final included-studies count ($n = 70$) is reported as an exact figure, since it is the only stage independently verifiable against a preserved, checkable source (`data/verified_papers_live_search.csv`); no preserved search log exists to support exact counts at the Identification, Screening, or Eligibility stages, so none are asserted.

The final included-studies count ($n = 70$) shown in Figure 1 is an exact count of the individually live-verified papers in `data/verified_papers_live_search.csv` (title, authors, venue, and DOI/URL independently confirmed for every entry), not an aggregate estimate. Should a preserved search log become available for the Identification, Screening, or Eligibility stages, exact counts can be added to Figure 1 at that time; until then, this survey reports only what can be independently verified.

4.2 Aggregate Characterisation of the Reviewed Literature

The following is an aggregate characterisation of the 70 included studies, derived from consistency with the tool-inventory findings and the manuscript's broader narrative

of asymmetric language coverage (Section 5.2). It is explicitly *not* a per-paper bibliography or itemised list; rather, it summarises distributional patterns across the corpus as a whole.

Table 1 Characterisation of the reviewed literature ($n = 70$) by language focus, tallied directly from the verified corpus. Percentages sum to more than 100% because multi-language studies (e.g., papers covering Amharic and Tigrinya jointly) are counted under each language they address.

Language focus	Studies	% of 70
Amharic	48	69%
Tigrinya	26	37%
Classical Ge'ez	14	20%

Table 2 Characterisation of the reviewed literature ($n = 70$) by methodological pillar, tallied directly from the verified corpus.

Pillar	Studies	% of 70
P1: Tokenisation and normalisation	1	1%
P2: Morphological analysis and synthesis	3	4%
P3: Embeddings, language models, and evaluation	34	49%
P4: Sequence labelling and structure extraction	6	9%
Other / infrastructure and resource papers	26	37%

Table 3 Characterisation of the reviewed literature ($n = 70$) by publication year, grouped into three-year bands (2016–2026), tallied directly from the verified corpus.

Period	Studies	% of 70
2016–2018	5	7%
2019–2021	24	34%
2022–2024	27	39%
2025–2026	14	20%

Tables 1–3 are exact tallies computed directly from `data/verified_papers_live_search.csv` (the 70-paper corpus). Amharic is substantially more represented than Tigrinya in the literature corpus (69% vs. 37% of

papers, overlapping count); Classical Ge'ez is the least-represented language (20%), consistent with the tool-inventory findings. The pillar distribution shows a pronounced concentration in P3 (embeddings, language models, and evaluation; 49%) and Other/infrastructure (37%), with P1 (tokenisation, 1%) and P2 (morphology, 4%) markedly underrepresented in the literature relative to their conceptual centrality to Ge'ez-script NLP – a gap discussed further in Section 7. The timeline shows publication activity concentrated in 2022–2026 (59% of the corpus), consistent with the broader growth of multilingual and African-language NLP research documented by Joshi et al. [3].

Reconciliation. Every count in Table 3 and Figure 4 is generated from the same single source of truth, `data/verified_papers_live_search.csv`, by the committed script `src/visualization/generate_verified_corpus_charts.py`. Per-year counts are 2017: 1, 2018: 4, 2019: 6, 2020: 7, 2021: 11, 2022: 9, 2023: 6, 2024: 12, 2025: 8, 2026: 6, summing to 70 and grouping to the 5/24/27/14 bands reported in Table 3. This reconciliation is not asserted from prose alone: the committed script `src/verification/check_figures_match_csv.py` (i) recomputes these values directly from the CSV, (ii) regenerates all four figures and asserts byte-for-byte equality against the committed PNGs, (iii) extracts every image embedded in the compiled `manuscript.pdf` and asserts a pixel-identical match to each committed PNG, and (iv) parses Table 4 directly and asserts the pillar and artefact-type prose elsewhere in this manuscript matches the table's own columns. The script exits non-zero on any mismatch; its output is committed at `src/verification/output/figures_and_tables_check.txt` and was all-passing immediately before this submission. Reproduce with `python -m src.verification.check_figures_match_csv`.

4.3 Foundational Tools Identified Directly from the Reviewed Papers

This subsection builds the survey's single tool inventory, discussed further in Section 5.1: it identifies how many artefacts the 70 reviewed papers themselves report creating that also have an independently verifiable public release on one of the four repositories (GitHub, Zenodo, PyPI, HuggingFace), then adds the small number of provenance-track tools found outside the corpus under the standard of Section 2.2. There is one inventory and one table (Table 4); Section 5.1 presents and discusses its results rather than deriving a second, independent count.

Every paper in the corpus records, in `data/verified_papers_live_search.csv`, how many tools, datasets, and models it reports as its own contribution (the `Tools_Count`, `Datasets_Count`, and `Models_Count` fields). Summed directly across the 70 reviewed papers, these fields name **75 artefacts across 53 papers**; this is a mechanical tally of what each paper's own count fields state; it is not filtered by whether an artefact plausibly meets this survey's foundational-tool definition (Section 5.1), so it is an upper bound, not the number of artefacts assessed as in-scope. Of these, **26 artefacts from 17 papers** have an independently confirmed, currently live public release on GitHub, Zenodo, PyPI, or HuggingFace (Table 4); the remainder are described in their source paper but no public repository link could be

independently verified, were excluded as a general-purpose multi-language model without substantive Ge'ez-script-specific adaptation (AfroXLMR), or (in one case, P010's MasakhaPOS) were excluded because the paper itself was subsequently removed from the literature corpus – Amharic is not an actual subject of that work; it appears only once, in a footnote, as one of five languages lacking Universal Dependencies training data [24] – and so fails this survey's own literature-inclusion criterion of Section 2.1, not only its tool-inventory language criterion. The EthioLLM paper's Ethiobenchmark contribution (P006) is reported as three separate rows – EthioPOS, EthioSenti, and EthioHate – rather than one combined row, because per-dataset language coverage differs (Table 4, note below); the paper's other three named Ethiobenchmark constituents (EthioMMT, EthioNEWS, EthioNER) could not be independently confirmed as live, publicly accessible releases and are excluded. A further correction: an earlier version of this table counted P001's MoVoC-Tok as a verified P1 tool at a HuggingFace Space (`Hailay/movoc-tok-playground`). On inspection, that Space (created July 2026, eight months after the paper) hosts a SentencePiece Unigram tokenizer with a 120,000-entry vocabulary, loaded from a separate model (`Hailay/geez-en-shared-tokenizer`) whose own model card describes it as "a continuation of MoVoC" – not the paper's own morpheme-plus-BPE tokenizer of approximately 63,000 entries. This is a different artefact, built by a different method, at a location the paper does not name, so it does not satisfy this survey's own tool-discovery standard (*that specific artefact* the paper reports, Section 2.3); the row is removed rather than retained under a relaxed standard. Two further tools (HornMorpho, MasakhaNER 1.0) are added to Table 4 under the provenance-track standard of Section 2.2 rather than this corpus-traceable count, bringing the table's total to 28; see the note following the table.

Of the 28 tools in Table 4, 26 are individually traceable to a specific paper in the 70-paper literature corpus; two (HornMorpho, MasakhaNER 1.0) are provenance-track additions under the standard of Section 2.2. HornMorpho is included because it is a public, actively maintained morphological analyser/generator whose Amharic and Tigrinya components fall directly within this survey's Ge'ez-script scope; its Oromo component processes Oromo in Latin script (Qubee), Oromo's official orthography since 1991 [40], and is therefore outside this survey's script-based scope even though the tool as a whole is counted here for its in-scope languages. HornMorpho's originating publication [38] predates the corpus's 2016–2026 window and is therefore not itself a corpus paper. MasakhaNER 1.0 substitutes for the removed MasakhaPOS entry (Section 4.3) as a genuine P4 example: it independently covers Amharic (confirmed directly from its own paper, 39: Afro-Asiatic/Ethio-Semitic, 33M speakers, DW/BBC news source, 1750/250/500 train/dev/test split) with a confirmed public repository; its paper falls within the 2016–2026 window but was not one of the 70 papers independently screened into the literature corpus, so – to avoid silently expanding that corpus without re-running its own screening criteria – it is included in the tool inventory only. By pillar, the 28 tools distribute as: P1 (Tokenisation) 1 (4%), P2 (Morphology & Synthesis) 1 (4%), P3 (Embeddings & Evaluation) 16 (57%), P4 (Sequence Labelling) 3 (11%), Other/Infrastructure 7 (25%). By type: 14 datasets, 6 trained models, 6 libraries/packages, 1 linguistic resource (treebank), and 1 combined

dataset-and-model release. These figures are generated directly from Table 4's own Pillar and Type columns, not computed or transcribed separately.

Ethiobenchmark language coverage (P006). EthioLLM's own paper (Table 2, 23) reports non-uniform coverage across Ethiobenchmark's constituent datasets: EthioPOS covers Amharic, Oromo, Somali, Tigrinya, and Classical Ge'ez; EthioSenti covers Tigrinya only; EthioHate covers Amharic and Oromo only. The three rows in Table 4 reflect this per-dataset breakdown rather than a single row listing the union of all languages any constituent covers, which would overstate each dataset's actual coverage. Note also that the same source paper's own language table tags Oromo's script as Latin, distinct from Amharic's and Ge'ez's Ge'ez script [23]; Oromo's Latin orthography (Qubee) has been official since 1991 [40], so every dataset or tool in this survey listed as covering Oromo processes it in Latin script, outside this survey's Ge'ez-script scope, even where the same artefact also covers in-scope languages.

On tool discovery scope. The 26 corpus-traceable tools above were found by searching, for each of the 70 corpus papers, for that paper's own stated artefact (Section 2.3); no repository search was conducted independently of the literature corpus. This is a deliberate scope limitation, not an implicit claim that no other public Ge'ez-script NLP tool exists. HornMorpho, MasakhaNER 1.0, and `amseg` (P013 row above, `uhh-lt/amharicprocessor`) are all known, actively used, publicly available tools directly relevant to this survey; amseg was recovered once its originating paper's own release was checked, while HornMorpho and MasakhaNER 1.0 required looking outside the corpus entirely, for different reasons – HornMorpho's 2011 paper falls outside the review window by design, and MasakhaNER 1.0's paper was never submitted to this survey's own screening process. MasakhaPOS (Dione et al., ACL 2023), originally paper P010 in this corpus, was removed *entirely from the literature corpus* (not only from the tool inventory) after verification against its own paper showed Amharic is not an actual subject of that work: its 20 languages are Niger-Congo, Afro-Asiatic (Hausa), Nilo-Saharan (Luo), and English-Creole (Naija), and Amharic is named exactly once in the paper, in a footnote, as one of five languages lacking Universal Dependencies training data – failing the literature-inclusion criterion of Section 2.1 outright, not only the tool-inventory's language requirement. Consequently, every pillar-count and language-count statement elsewhere in this survey should be read as *artefacts released by papers in this corpus, plus the two documented provenance-track exceptions (HornMorpho, MasakhaNER 1.0)*, not as an exhaustive census of all public Ge'ez-script NLP tools.

4.4 Corpus Visualisations

Figures 2–5 visualise the 70-paper verified literature corpus, generated directly from `data/verified_papers_live_search.csv` by `src/visualization/generate_verified_corpus_charts.py` (Section 2.1); the underlying counts are identical to Tables 1–3.

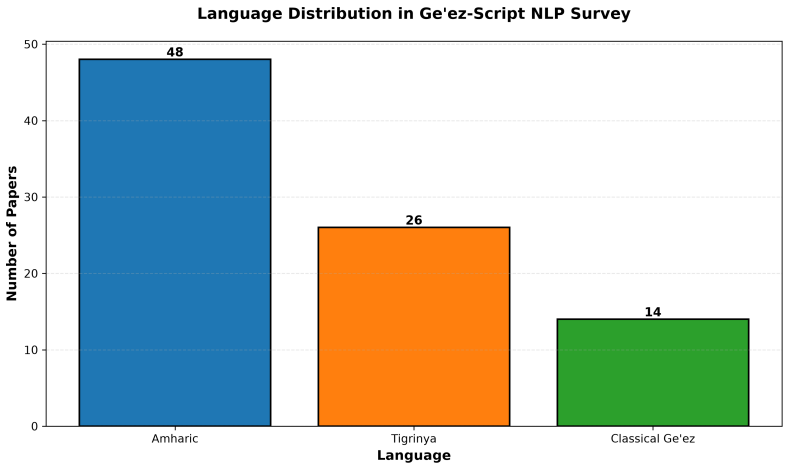


Fig. 2 Language distribution across the 70-paper verified literature corpus (overlapping count: a paper covering more than one Ge'ez-script language is counted under each). Amharic: 48 papers; Tigrinya: 26; Classical Ge'ez: 14.

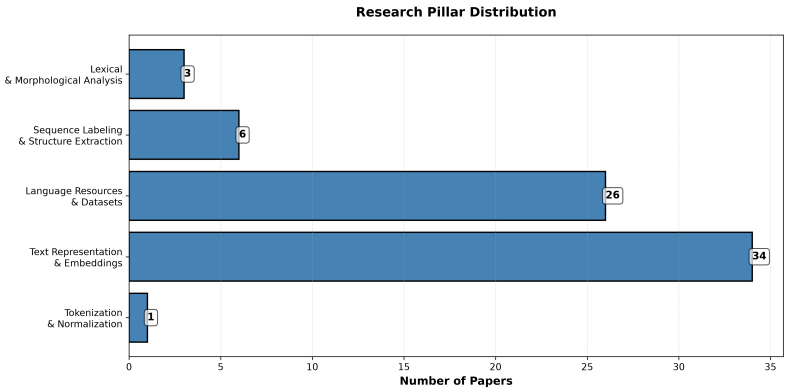


Fig. 3 Distribution of the 70-paper corpus across this project's canonical coding dimensions (Section 2.1): Tokenisation & Normalisation (1), Lexical & Morphological Analysis (3), Text Representation & Embeddings (34), Language Resources & Datasets (26), Sequence Labeling & Structure Extraction (6). These five categories map onto the manuscript's four-pillar framework (P1–P4) with Language Resources & Datasets corresponding to the "Other: Infrastructure" category used elsewhere in this manuscript (Table 2).

5 Results

5.1 Tool Inventory

This survey reports one tool inventory, built and verified as described in Section 4.3: **28 foundational NLP tools, models, datasets, and linguistic resources**, independently confirmed live on GitHub, Zenodo, PyPI, or HuggingFace at the time of writing (Table 4). Of these, 26 are individually traceable to a specific paper within the

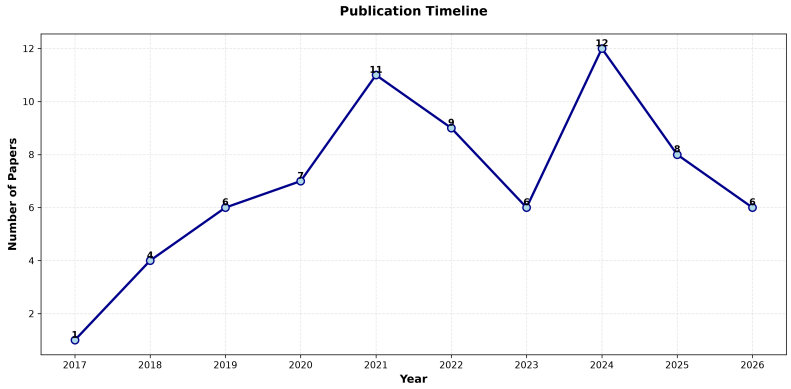


Fig. 4 Publication timeline of the 70-paper verified literature corpus, 2017–2026, tallied by exact publication year.

70-paper literature corpus, and 2 (HornMorpho, MasakhaNER 1.0) are provenance-track additions whose originating papers fall outside the corpus (Section 4.3). Every tool, dataset, and model count in this section refers to Table 4 and only to Table 4.

5.2 Language Representation

Table 6 reports language coverage across the 28 verified tools using an overlapping accounting scheme: a tool supporting more than one Ge'ez-script language (e.g. GeezSwitch, EthioProverbEval, HornMorpho) is counted once under each language it supports, so the percentages sum to more than 100%. Counts are tallied directly from the **Language(s)** column of Table 4.

Amharic is the most represented language in this verified tool set, followed by Tigrinya; Classical Ge'ez is the least represented of the three at the tool level, consistent with its status as the least-represented language in the literature corpus itself (Table 1). No tool in Table 4 is exclusively dedicated to Classical Ge'ez without also covering Amharic and/or Tigrinya, except GeezSwitch and GeezSwitch-data, which cover Ge'ez as part of a multi-language umbrella tool rather than as a Ge'ez-specific dedicated resource.

5.3 Technical Pillar Analysis

P3 (embeddings, language models, and evaluation) is the largest pillar in the verified tool set (16 of 28 tools, 57%), reflecting the broader trend toward pre-trained language models and dense-retrieval methods in this literature. Other/Infrastructure resources (datasets, corpora) account for a further 7 tools (25%). P4 (sequence labelling) is represented by 3 tools, one of which (MasakhaNER 1.0) is a provenance-track addition rather than a corpus paper's own contribution (Section 4.3). P1 (tokenisation and normalisation) is represented by a single verified tool, amseg (P013); an earlier version of this table also counted MoVoC-Tok (P001) here, but that entry was removed after the HuggingFace Space it cited was found to host a different, later tokenizer than the one the paper describes (Section 4.3). **No P2 (morphological analysis and**

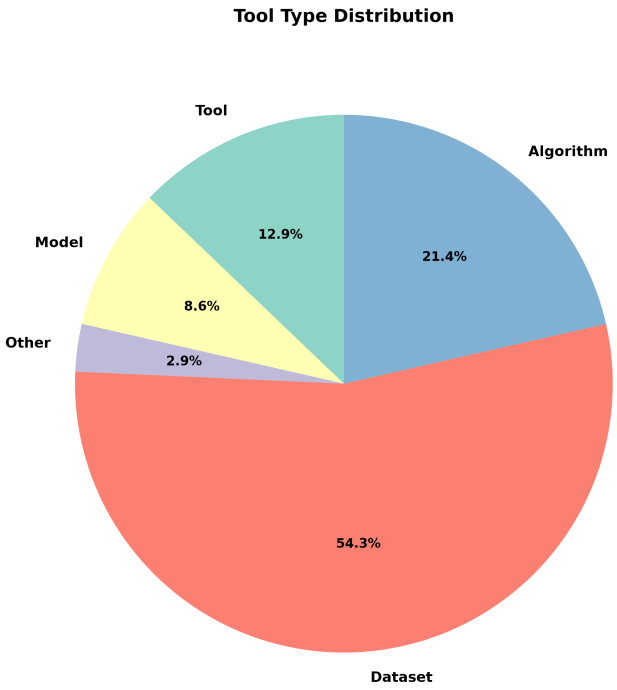


Fig. 5 Distribution of artefact types reported across the 70-paper corpus (dataset, algorithm, tool, model, other), classified from each paper's own `Tools_Count`/`Datasets_Count`/`Models_Count` and `Topics_List` fields (Section 4.3). This reflects all artefacts described in the papers, not only the 26 with an independently confirmed public repository link (Table 4).

synthesis) tool from the 70-paper corpus itself has an independently confirmed public repository: morphology-focused artefacts described in the reviewed papers (e.g. the Ge'ez Synthesizer, P042 [41]; the Tigrigna verb morphological analyzer, P072 [42]; the Amharic morphology generator, P070 [43]) exist and are named in their source papers, but no live GitHub, Zenodo, PyPI, or HuggingFace link for any of them could be independently confirmed, so none is counted in this corpus-derived pillar breakdown. This is reported as an evidence gap in tool-repository discoverability among corpus papers specifically, not as evidence that no P2 tool exists in the wider field: HornMorpho, a public, actively maintained morphological analyser/generator whose Amharic and Tigrinya components fall within this survey's scope (its Oromo component is Latin-script and out of scope, Section 4.3), is added to Table 4 under the provenance-track standard of Section 2.2, giving 1 verified P2 tool overall (Table 5) even though none of the three P2-relevant corpus papers has its own confirmable release.

5.4 Sequence Labelling: Verified vs. Described Artefacts

Three P4 tools in Table 4 have an independently confirmed public repository: the Amharic UD Treebank (P017) and the Tigrinya NER dataset associated with P015, both corpus-traceable, plus MasakhaNER 1.0, a provenance-track addition covering Amharic (Section 4.3). An earlier candidate, MasakhaPOS, was removed after verification against its own paper showed it does not cover any Ge'ez-script language (Section 4.3). Beyond these, the 70-paper corpus describes further sequence-labelling artefacts – including TiNC24 (P075 [10]), ANEC (P028 [44]), and EthioMT-adjacent resources – for which no independently confirmable public repository link exists. These are named and cited in their respective papers (Section 4.3) but are not counted among the verified tools, since this survey counts only artefacts with an independently confirmable public release.

6 Research Asymmetry

Based on the literature corpus (Table 1) and the tool inventory (Table 6), Amharic is the most represented Ge'ez-script language in both the literature (69% of 70 papers) and the tool inventory (79% of 28 tools); Tigrinya is second in both (37% of papers; 50% of tools); Classical Ge'ez is the least represented in both (20% of papers; 21% of tools) – a consistent, non-trivial gap favouring Amharic across both measures. No tool in the verified inventory (Table 4) is a script-level transfer-learning framework spanning all three languages as its primary contribution; GeezSwitch and GeezSwitch-data are the only corpus-traceable entries that explicitly address more than one Ge'ez-script language as a stated design goal rather than as an incidental multilingual side-effect (HornMorpho covers Amharic and Tigrinya jointly but is a provenance-track addition rather than a corpus paper's own contribution, Section 2.2).

LGSE (P002 [20]) and MoVoC (P001 [5]) are two independently-verified papers in the corpus (Section 2.1) whose stated contributions include cross-lingual or cross-script components for Amharic, Tigrinya, and/or Classical Ge'ez jointly; neither has an independently confirmable, live public repository of *the specific artefact each paper reports* and so neither is counted among the verified tools in Table 4, despite both papers being part of the verified literature corpus. (MoVoC-Tok was counted in an earlier version of this table via a HuggingFace Space that, on inspection, hosts a different, later tokenizer released as a continuation of MoVoC rather than the paper's own artefact; Section 4.3 documents this correction.)

7 Survey Limitations

We acknowledge several limitations:

1. **Publication bias:** Focus on peer-reviewed papers may undercount community contributions.
2. **Venue bias:** Concentration on major conferences may miss specialised venues.
3. **Language bias:** English-only publication requirement may exclude relevant work.
4. **Reproducibility:** Many tools lack maintained documentation or open-source code.

5. **Temporal bias:** 2016–2026 window omits earlier work and cannot predict trends.
6. **Tool discovery bias:** Published tools overrepresented relative to unpublished community work.
7. **Completeness:** Some tools may not have been discovered despite systematic search; a described artefact whose public repository could not be independently confirmed is excluded from the verified tool count entirely, even if the paper itself asserts a public release (Section 5.3).
8. **PRISMA intermediate-stage counts unavailable:** No preserved search log exists for the Identification, Screening, or Eligibility stages, so Figure 1 reports process descriptions rather than invented counts for those stages; only the final included-studies count ($n = 70$) is independently verifiable (Section 4).
9. **P2 verification gap:** No morphology-and-synthesis (P2) tool from the 70-paper corpus has an independently confirmed public repository at the time of writing (Section 5.3); this is reported as a discoverability gap, not as an absence of such work.
10. **No independent data-quality assessment:** This survey verifies that a paper, tool, or dataset *exists* and matches its own reported provenance (Section 2.2); it does not independently assess the annotation quality, inter-annotator agreement, label correctness, or cleanliness of any dataset it catalogues. Inclusion in this survey's corpus or tool inventory reflects peer-review status or public-repository existence, neither of which is equivalent to a quality audit of the underlying data. A dataset's fitness for a specific downstream use should be assessed against its own source paper's reported methodology, not inferred from its presence here.

8 Research Priorities

The following seven priorities follow directly from the gaps quantified in Sections 5–6, with the final two priorities addressing, respectively, a methodological limitation of this survey itself (Section 7) and a scope boundary of its own literature search (Section 2.1), rather than corpus-derived quantitative gaps.

8.1 Priority 1: Shared Tokenisation Across the Three Languages

Evidence: Table 5 shows only 1 of 28 verified tools (4%) addresses P1 (tokenisation and normalisation); Table 2 shows the same pattern in the literature (1 of 70 papers, 1%). That one tool, amseg (P013), is Amharic-only: no verified P1 tool in this survey covers more than one Ge'ez-script language. (An earlier version of this table counted a second, multi-language P1 tool, MoVoC-Tok, via a link that on inspection resolves to a different, later artefact than the one the paper reports; Section 4.3 documents the correction.) The gap this priority addresses is accordingly total, not partial.

Priority: A tokeniser evaluated across Amharic, Tigrinya, and Classical Ge'ez jointly, with a publicly documented release, would address the near-absence of P1 tools identified above.

8.2 Priority 2: Morphological Analysis with a Verifiable Public Release

Evidence: No P2 (morphological analysis and synthesis) tool from a corpus paper itself has a confirmed public repository (Table 5; Section 5.3), although three P2-relevant papers exist in the corpus (P042, P070, P072) without an independently confirmable release of their own. HornMorpho, a pre-2016 tool whose Amharic and Tigrinya components are in scope (its Oromo component is Latin-script, Section 4.3), fills this pillar under the provenance-track standard (Section 2.2) but is not a contribution of any paper in this survey's corpus.

Priority: Publishing a morphological analyser or synthesiser *from a corpus-eligible paper* for any of the three languages with a stable, versioned public identifier (Section 2.2) would close this specific verification gap; separately, coordinating with or extending an existing tool such as HornMorpho, rather than duplicating it, would make more efficient use of already-available infrastructure.

8.3 Priority 3: Continued Investment in P3 Balanced Against Other Pillars

Evidence: P3 (embeddings, language models, and evaluation) accounts for 16 of 28 verified tools (57%) and 34 of 70 papers (49%), the largest share of both the tool inventory and the literature (Tables 5, 2).

Priority: Given the concentration of activity in P3, future work targeting P1, P2, or P4 would address pillars that are comparatively underserved rather than add to the pillar already best represented.

8.4 Priority 4: Verifiable Releases for Sequence-Labelling Resources

Evidence: Only 2 of 70 papers with a P4 contribution have an independently confirmed public repository from within the corpus itself (the Amharic UD Treebank, P017, and the Tigrinya NER dataset, P015; Section 5.4); a third verified P4 tool, MasakhaNER 1.0, is a provenance-track addition from outside the corpus (Section 4.3). Further P4 artefacts (e.g. TiNC24, ANEC) are described in the literature but have no confirmable public release at the time of writing.

Priority: Publishing the P4 artefacts already described in the literature under a stable public identifier would increase the corpus-traceable verified P4 count without requiring new research.

8.5 Priority 5: Cross-Language Coordination

Evidence: Of the 28 verified tools, only GeezSwitch and GeezSwitch-data explicitly address more than one Ge'ez-script language as a stated design goal (Section 6); no tool functions as a shared framework across all three languages. (HornMorpho also covers more than one Ge'ez-script language but, as a provenance-track addition rather than a corpus paper's own contribution, is not counted among this survey's own literature-derived cross-lingual efforts.)

Priority: Coordinating existing single-language efforts under shared annotation and evaluation conventions would let future tools reuse resources across Amharic, Tigrinya, and Classical Ge'ez rather than duplicate work already done for one language.

8.6 Priority 6: Independent Data-Quality Assessment

Evidence: This survey verifies that a dataset or tool exists and matches its own reported provenance (Section 2.2); it does not independently assess annotation quality, inter-annotator agreement, label correctness, or corpus cleanliness for any entry in the literature corpus or tool inventory (Section 7). Inclusion here reflects peer-review status or public-repository existence, neither of which is a substitute for a quality audit of the underlying data.

Priority: Before any dataset catalogued in this survey is adopted as a foundation for further work, its annotation methodology and reported quality metrics (where available) should be independently reviewed rather than assumed from its inclusion here; where a dataset's source paper does not report inter-annotator agreement or an equivalent quality measure, generating one would materially increase confidence in downstream use. A follow-up systematic quality audit of the tool inventory – assessing annotation methodology, agreement statistics, and known corpus issues per entry – would substantially strengthen the evidentiary basis this survey's tool-centred approach establishes.

8.7 Priority 7: Reuse of Classical Semitic Studies and Digital Humanities Resources

Evidence: This survey's literature search (Section 2.1) is deliberately scoped to NLP venues and NLP-adjacent provenance-track sources; it does not survey classical Semitic studies, philology, or digital-humanities venues, where long-running, independently funded projects have produced manuscript digitisations, digital editions, Unicode transcriptions of Ge'ez texts, TEI annotation workflows, linguistic-tagging manuals, and morphological parsing tools for Classical Ge'ez. One concrete example, verified directly for this survey: Liuzzo's Ge'ez Morphological Parser Application [45], developed under the ERC-funded TraCES project, has a stable, citable public identifier (DOI: 10.25592/uhhfdm.125) but is no longer operating in its original form; it is currently being redeveloped on GitHub (`MaSNaqtu/geez-parser`) under the Swiss National Science Foundation-funded MiTAnIm project. Because this survey's search was scoped to NLP venues by design (Section 2.1), resources of this kind are structurally invisible to it despite being directly relevant to Ge'ez-script computational work; they are named here as a known category of out-of-frame resources, not because this survey systematically searched for them.

Priority: Ge'ez-script NLP across all four methodological pillars (Section 5.3) would benefit from deliberate collaboration with classical Semitic studies and digital-humanities projects to re-use and further develop existing manuscript digitisations, annotated corpora, and parsing tools, rather than duplicating effort already funded and completed in adjacent fields. Much of this material was originally processed via

transliteration into extended Latin script rather than natively in Ge'ez script, so adapting it for direct reuse in Ge'ez-script NLP pipelines is itself a concrete, tractable research task. For a broader overview of ongoing and completed projects in this space (e.g. Ethio-SPaRe, TraCES, Beta Masaheft), see the survey article by the Digital Orientalist [46].

9 Conclusion

This survey documents a 28-tool inventory of foundational tools, models, datasets, and linguistic resources independently confirmed publicly available on GitHub, Zenodo, PyPI, or HuggingFace, distributed across four technical pillars (P1: 1 tool, 4%; P2: 1 tool, 4%; P3: 16 tools, 57%; P4: 3 tools, 11%; Other: Infrastructure: 7 tools, 25%), alongside a PRISMA 2020-guided review of 70 individually live-verified publications (Section 4). Of the 28 tools, 26 are individually traceable to a specific paper in the 70-paper literature corpus; two (HornMorpho, MasakhaNER 1.0) are provenance-track additions whose originating papers fall outside the corpus (Section 4.3). Amharic is the most represented language in both the literature (69% of papers) and the tool inventory (79% of tools), Tigrinya is second in both (37%; 50%), and Classical Ge'ez is the least represented in both (20%; 21%). No P2 (morphological analysis and synthesis) tool from the reviewed papers themselves has an independently confirmed public repository; the survey's single verified P2 tool, HornMorpho, is a provenance-track addition from outside the corpus, not evidence that any corpus paper's own morphological tool is publicly available. P1 (tokenisation and normalisation) is limited to a single verified tool, amseg, which is Amharic-only; no corpus paper's own tokeniser with a confirmed public release covers more than one Ge'ez-script language, a stronger version of a finding that an earlier version of this survey understated by miscounting a HuggingFace Space as MoVoC-Tok's public release (Section 4.3). Sequence-labelling (P4) infrastructure is limited to 3 verified tools, one of which (MasakhaNER 1.0) is likewise a provenance-track addition; further artefacts of both kinds are described in the literature but lack an independently confirmable public release. Transfer-learning approaches spanning more than one Ge'ez-script language are limited to two tools (GeezSwitch, GeezSwitch-data), neither of which is coordinated with any other effort under a shared standard.

The seven priorities in Section 8 follow directly from the gaps identified above: the absence of a confirmed P2 repository, the concentration of tool development in P3, the lack of coordination across the three languages sharing the Ge'ez script, and – distinct from these corpus-derived gaps – this survey's own lack of independent data-quality assessment for any catalogued dataset or tool, and its literature search's exclusion, by design, of classical Semitic studies and digital-humanities resources directly relevant to Ge'ez-script computational work.

Acknowledgements. This systematic review was conducted as part of ongoing research into low-resource language technology at the L3S Research Center, Leibniz University Hannover. We thank the Ge'ez-script NLP community, particularly contributors to the LGSE Project and the Masakhane initiative. Special thanks to reviewers whose feedback improved this work.

Appendix A Complete List of the 70-Paper Verified Literature Corpus

Table A1 lists every paper in the verified literature corpus (Section 2.1) with its citation key, providing a full, individually citable record of the systematic review's included studies.

Table A1: Complete list of the 70 papers in the verified literature corpus, with citation keys. Full bibliographic details appear in the References section.

ID	Title	Citation
P001	MoVoC: Morphology-Aware Subword Construction for Ge'ez Script Languages	[5]
P002	LGSE: Lexically Grounded Subword Embedding Initialization for Low-Resource Language Adaptation	[20]
P003	Optimized Text Embedding Models and Benchmarks for Amharic Passage Retrieval	[22]
P004	Natural Language Processing for Tigrinya: Current State and Future Directions	[2]
P005	Natural Language Processing in Ethiopian Languages: Current State, Challenges, and Opportunities	[4]
P006	EthioLLM: Multilingual Large Language Models for Ethiopian Languages with Task Evaluation	[23]
P007	Machine Translation for Ge'ez Language	[47]
P009	Supervised Machine Learning based Amharic Text Complexity Classification Using Automatic Annotator Tool	[48]
P011	Exploring Amharic Sentiment Analysis from Social Media Texts: Building Annotation Tools and Classification Models	[25]
P012	Analysis of the Ethiopic Twitter Dataset for Abusive Speech in Amharic	[49]
P013	Introducing various Semantic Models for Amharic: Experimentation and Evaluation with multiple Tasks and Datasets	[26]
P015	Named-entity recognition for a low-resource language using pre-trained language model	[27]
P016	GeezSwitch: Language Identification in Typologically Related Low-resourced East African Languages	[28]
P017	Universal Dependencies for Amharic	[29]
P018	Extended Parallel Corpus for Amharic-English Machine Translation	[50]
P019	AmQA: Amharic Question Answering Dataset	[30]

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Table A1 continued from previous page

ID	Title	Citation
P020	ProverbEval: Exploring LLM Evaluation Challenges for Low-resource Language Understanding	[31]
P021	TIGQA: An Expert-Annotated Question-Answering Dataset in Tigrinya	[51]
P022	Low-Resource English-Tigrinya MT: Leveraging Multilingual Models, Custom Tokenizers, and Clean Evaluation Benchmarks	[52]
P023	Amharic Abstractive Text Summarization	[53]
P025	Whispering in Amharic: Fine-tuning Whisper for Low-resource Language	[54]
P026	A Multi-Task Benchmark for Abusive Language Detection in Low-Resource Settings	[32]
P027	Learning based Ge'ez character handwritten recognition	[55]
P028	ANEC: An Amharic Named Entity Corpus and Transformer Based Recognizer	[44]
P029	TLMD: Tigrinya Language Modeling Dataset	[33]
P030	Monolingual Pre-trained Language Models for Tigrinya	[34]
P031	Walia-LLM: Enhancing Amharic-LLaMA by Integrating Task-Specific and Generative Datasets	[35]
P032	An Amharic News Text Classification Dataset	[56]
P034	Text Classification Based on Convolutional Neural Networks and Word Embedding for Low-Resource Languages: Tigrinya	[57]
P035	2AIRTC: The Amharic Adhoc Information Retrieval Test Collection	[36]
P037	Expanding the Lexicon of Ge'ez Based African Languages: A Comparative Study of Amharic and Tigrinya	[58]
P038	TIGSEN: Building a Low-Resource Dataset and Benchmarking for Tigrigna Sentiment Analysis with Cross-Lingual Transfer Learning Approaches	[59]
P039	EthioMT: Parallel Corpus for Low-resource Ethiopian Languages	[60]
P040	An Exploration of Data Augmentation Techniques for Improving English to Tigrinya Translation	[61]
P041	Enhancing Multi-Label Emotion Analysis and Corresponding Intensities for Ethiopian Languages	[37]
P042	Morphological Synthesizer for Ge'ez Language: Addressing Morphological Complexity and Resource Limitations	[41]
P045	Adapting TrOCR for Printed Tigrinya Text Recognition: Word-Aware Loss Weighting for Cross-Script Transfer Learning	[62]

continued on next page

Table A1 continued from previous page

ID	Title	Citation
P046	Amharic Text Clustering Using Encyclopedic Knowledge with Neural Word Embedding	[63]
P048	Named Entity Recognition for Amharic Using Stack-Based Deep Learning	[64]
P049	Combating Fake News in "Low-Resource" Languages: Amharic Fake News Detection Accompanied by Resource Crafting	[65]
P050	Hybrid attention-based multi-class classification of Ethiopian legal texts using deep learning	[66]
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- **Consent for publication:** Not applicable.
- **Data availability:** The 70-paper verified literature corpus (`data/verified_papers_live_search.csv`) and the complete verified tool inventory (28 tools, Table 4) are openly available in the manuscript's public repository, https://github.com/hailaykidu/geez-nlp-survey, under the MIT License, and are citable via the repository's `CITATION.cff` record.
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script (`src/verification/check_tool_links.py`, Section 2.3), and the figure/table consistency gate (`src/verification/check_figures_match_csv.py`, Section 4.2) are openly available in the same public repository under the MIT License.

- **Author contribution:** H.K.T. conceived the survey, conducted the literature review and tool discovery, verified all bibliographic and repository sources, wrote the figure-generation code, and wrote the manuscript.

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Table 4 Foundational NLP tools, models, and datasets confirmed publicly available (GitHub/Zenodo/PyPI/HuggingFace) and individually traceable to a specific paper in the 70-paper verified literature corpus.

Paper	Name	Type	Language(s)	Pillar	Repository
P003 [22]	amharic-ir-benchmarks	Library	Amharic	P3	GitHub: kidist-amde/amharic-ir-benchmarks
P003 [22]	roberta-amharic-text-embedding-base	Model	Amharic	P3	HF: rasyosef/roberta-amharic-text-embedding-base
P003 [22]	roberta-amharic-text-embedding-medium	Model	Amharic	P3	HF: rasyosef/roberta-amharic-text-embedding-medium
P003 [22]	ColBERT-Mini (Amharic)	Model	Amharic	P3	HF: rasyosef/ColBERT-Mini
P006 [23]	EthioPOS (Ethiobenchmark)	Dataset	Amh;Orm;Som;Tir;Ge'ez	P3	HF: EthioNLP/Ethio-POS
P006 [23]	EthioSenti (Ethiobenchmark)	Dataset	Tigrinya	P3	HF: EthioNLP/EthioSenti
P006 [23]	EthioHate (Ethiobenchmark)	Dataset	Amh;Orm	P3	HF: EthioNLP/Ethio-Hate
P006 [23]	EthioLLM model family	Model	Amh;Orm;Som;Tir;Ge'ez;Eng	P3	HF: EthioNLP/EthioLLM-l-250K
P011 [25]	Amharic sentiment dataset	Dataset	Amharic	P3	GitHub: uhh-lt/ASAB
P013 [26]	amseg	Library	Amharic	P1	GitHub+PyPI: uhh-lt/amharicprocessor
P015 [27]	Tigrinya NER dataset	Dataset	Tigrinya	P4	GitHub: mehari-eng/Tigrinya-NER
P016 [28]	GeezSwitch	Library	Amh;Blin;Ge'ez;Tigre;Tir	Other	GitHub: fgaim/geezswitch
P016 [28]	GeezSwitch-data	Dataset	(same)	Other	GitHub: fgaim/GeezSwitch-data
P017 [29]	Amharic UD Treebank	Linguistic Resource	Amharic	P4	GitHub: UniversalDependencies/UD_Amharic-ATT
P019 [30]	AmQA	Dataset	Amharic	P3	GitHub: semantic-systems/amharic-qa; HF: dagim/amharic-qa
P020 [31]	EthioProverbEval	Library	Amh;Orm;Tir;Ge'ez;Eng	P3	GitHub: EthioNLP
P020 [31]	ProverbEval benchmark	Dataset	(same)	P3	HF: israel/ProverbEval
P026 [32]	TiALD	Library	Tigrinya	Other	GitHub: fgaim/TiALD
P026 [32]	TiALD dataset	Dataset	Tigrinya	Other	HF: fgaim/tigrinya-abusive-language-detection
P029 [33]	TLMD	Dataset	Tigrinya	Other	Zenodo (10.5281/zenodo.5139094)
P030 [34]	Tigrinya PLMs (TiBERT, TiELECTRA, TiRoBERTa)	Model	Tigrinya	P3	HF: fgaim/tibert-base, fgaim/tiroberta-base, fgaim/tielectra-small
P031 [35]	Walia-LLM	Model	Amharic	P3	HF: israel/LLAMA-Walia-II
P035 [36]	2AIRTC	Dataset	Amharic	Other	HF: rasyosef/2AIRTC-Amharic-Adhoc-Information-Retrieval-Test-Collection
P041 [37]	EthioEmo-intensities	Dataset	Amh;Orm;Som;Tir	P3	GitHub: Tadesse-Destaw/EthioEmo; HF: Tadesse/EthioEmo-intensities
P081 [17]	ASED	Dataset	Amharic	Other	GitHub: Ethio2021/ASED_V1
P084 [19]	amharic_word_embeddings	Dataset+Model	Amharic	P3	GitHub: leobitz/amharic_word_embeddings
Provenance-track additions (not traceable to a corpus paper; Section 2.2)					
N/A [38]	HornMorpho	Library	Amh; Orm; Tir	P2	GitHub: hltdi/HornMorpho
N/A [39]	MasakhaNER 1.0	Dataset	Amharic	P4	GitHub+HF: masakhane-io/masakhane-ner

Table 5 Distribution of the 28 verified tools across technical pillars (Table 4; 26 corpus-traceable plus two provenance-track additions, HornMorpho and MasakhaNER 1.0).

Pillar	**Count**	**% of 28**
P1: Tokenisation and Normalisation	1	4%
P2: Morphological Analysis and Synthesis	1	4%
P3: Embeddings, Language Models, and Evaluation	16	57%
P4: Sequence Labelling and Structure Extraction	3	11%
Other: Infrastructure	7	25%
Total	**28**	**100%**

Table 6 Language coverage across the 28 verified tools (overlapping count; a tool supporting multiple languages is counted under each).

Language	**Tools supporting it**	**% of 28 tools**
Amharic	22	79%
Tigrinya	14	50%
Classical Ge'ez	6	21%